Automating EC2 Backups with Python and Lambda

Automating EC2 Backups with Python and Lambda

Why I Automated My EC2 Backups

Backups are one of those tasks in cloud computing that are essential but easy to overlook. When you're managing multiple EC2 instances, remembering to manually create snapshots every day or before key deployments can become a repetitive and risky process.

Missing just one backup can lead to **data loss, extended downtime**, or **costly troubleshooting** — especially if something goes wrong and there's no recent snapshot to recover from.

This was the exact problem I faced while working with EC2 instances on a regular basis. I needed a way to:

- Ensure my instances were backed up regularly
- Avoid the hassle of doing it manually
- Add reliability without increasing complexity or cost

That's when I turned to **Python and AWS Lambda** — two powerful tools that allowed me to create a lightweight, serverless automation system for EC2 backups.

What You'll Learn

In this blog, you'll learn how to automate EC2 backups on AWS using a simple and reliable serverless setup.
By the end of this guide, you'll be able to:

- ✅ Automatically create snapshots (AMIs) of your EC2 instances.
- ✅ Write a Python script using Boto3 (AWS SDK for Python)

- ✅ Deploy that script inside an AWS Lambda function.
- ✅ Use CloudWatch Events to schedule your backups (daily, weekly, etc.)
- ✅ Tag EC2 instances to control which ones get backed up — no need to hardcode anything.
- ✅ Monitor and test your setup using Lambda logs and the EC2 console.

Service Lineup

🖥️ Amazon EC2 (Elastic Compute Cloud)

EC2 provides virtual machines (instances) that run your applications, servers, or any compute workload in the cloud. These are the resources we'll back up.

⚙️ AWS Lambda

Lambda is a serverless compute service that runs your Python code without provisioning any servers. We'll use it to automate the backup logic.

⏰ Amazon CloudWatch Events

CloudWatch lets you schedule events (like a cron job) to trigger your Lambda function automatically — daily, weekly, or as you prefer.

🔐 AWS IAM (Identity and Access Management)

IAM controls permissions and security. We'll create an IAM role that allows our Lambda function to access and manage EC2 instances safely.

Read the full step-by-step guide on my website.

cloudjiva.com

CLOUD JIVA